

On the Gauge Nature of Time-Dependent Metrics

A Critique of the Space Dilation Model of Gravity

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Abstract

The Space Dilation model proposes that gravity arises from the exponential expansion of local spatial meters near matter. This paper evaluates the model's consistency through three complementary lenses. Kinematically, we show that the model implies an observer bifurcation: comoving observers are in free fall (0 g), while stationary surface observers experience no spatial metric change but must constantly move through the coordinate grid. Geometrically, we demonstrate that interpreting the model as a coordinate re-description violates general covariance, as dropping the required off-diagonal metric cross-terms alters the curvature of the spacetime. Empirically, we establish that if the spatial expansion were physical, the resulting drift in the measured gravitational constant would exceed precision Lunar Laser Ranging limits by 18 orders of magnitude. These results suggest that the physical source of surface proper acceleration is the static lapse gradient rather than a dynamic spatial metric evolution.

1. Introduction

The Space Dilation model, as presented at the International Summit on Gravitation, Astrophysics and Cosmology (ISGAC 2026; Levitt, 2026b) and as formalized in the primary text (Levitt, 2026a), rests on three related but distinct claims:

1. **A Numerical Observation:** The Newtonian surface acceleration:

$$g = \frac{GM}{R^2}$$

(where G is Newton's gravitational constant, M is the mass of the spherical body, and R is its radius) can be recovered by twice-differentiating an exponential scale factor $e^{t/T}$ where:

$$T = \sqrt{\frac{R^3}{GM}}$$

suggesting that local spatial meters double in proper length every $T \ln 2 \approx 558$ s at Earth's surface (approximately 9.3 minutes; Levitt, 2026b), corresponding to a characteristic e-folding timescale of $T \approx 806$ s (Levitt, 2026b). This characteristic e-folding timescale T is the strict inverse of the orbital angular frequency $\omega = \sqrt{GM/R^3}$ for a hypothetical, surface-skimming circular test particle ($\omega = 1/T$). This maps directly to Kepler's Third Law, where the square of the orbital period $\tau_{\text{orbit}} = 2\pi/\omega = 2\pi T$ scales proportionally with R^3 via $\tau_{\text{orbit}}^2 = \frac{4\pi^2}{GM} R^3$. Here, we distinguish between the linear scale factor $e^{t/T}$ governing physical distance intervals and the quadratic factor $e^{2t/T}$ that appears in the spatial components of the metric tensor (see Definition 1). Twice-differentiating the proposed exponential scale factor $e^{t/T}$ with respect to coordinate time t and evaluating at $t = 0$ yields:

$$\left. \frac{d^2}{dt^2} e^{t/T} \right|_{t=0} = \frac{1}{T^2}$$

Multiplying this result by the radius R naturally outputs:

$$\frac{R}{T^2} = \omega^2 R = \frac{GM}{R^2} = g$$

which is identical to the Newtonian centripetal acceleration $a_c = v^2/R = \omega^2 R$ (where $v = \omega R$ is the orbital velocity of the test particle). Consequently, the relation $R/T^2 = g$ is an algebraic tautology derived directly from Kepler's Third Law and the definition of centripetal acceleration, which is mathematically indistinguishable from classical mechanics. By defining the timescale T explicitly in terms of GM and R , the model utilizes T as a notational placeholder rather than a predictive physical parameter. Observe that the angular velocity identity $\omega = 1/T$ holds only at the specific radius $r = R$. Defining a single, radius-dependent parameter T and applying it globally as a constant timescale in a homogeneous scale factor $e^{t/T}$ is mathematically inconsistent with a radially varying gravitational field. In any realistic physical geometry, T must vary with radius ($T(r)$), which would introduce spatial gradients $(\partial_r T)$ that the diagonal, homogeneous model ignores.

2. **A Physical Assertion:** The primary text describes space as undergoing a *volume acceleration* of $4\pi GM \text{ m}^3/\text{s}^2$ emerging outward from matter, characterizing this outward stretching of volumes of space as the physical mechanism responsible for gravity (Levitt, 2026a; 2026b). However, this physical assertion is mathematically inconsistent with the proposed metric scale factor $e^{t/T}$. As shown in Section 4.1, a physical comoving expansion under this metric produces a time-dependent volume acceleration that is three times larger at the surface ($12\pi GM$), whereas enforcing a constant volume acceleration forces a power-law expansion ($r \propto t^{2/3}$) that contradicts the exponential ansatz.
3. **A Reframing Defense:** The model presents its exponential spatial expansion framework as a simpler, more intuitive description that is fully consistent with and derived from Einstein's general relativity, introducing no new physics and no inconsistent content (Levitt, 2026a; Levitt, 2026b).

The first claim is correct and mathematically uncontroversial. The second and third claims are what this paper addresses. The primary text defends this framework by asserting that the model is simply a re-parametrization of standard General Relativity with respect to proper time rather than coordinate time (Levitt, 2026a; 2026b). The thesis claims that as long as the differentiation is performed with respect to proper time, the underlying mathematical framework of Einsteinian physics remains completely unaltered.

Throughout this paper, we work in standard physical units where the speed of light c is kept explicit to facilitate direct comparison with empirical baseline measurements.

A mathematically rigorous delineation between a valid coordinate re-parametrization (a gauge choice) and a modification of the physical geometry is essential.

Definition 1 (Gauge Transformation vs. Non-Diffeomorphic Metric Alteration). *A valid "gauge transformation" in General Relativity is defined as a smooth coordinate chart transition governed strictly by diffeomorphism invariance on the spacetime manifold ($x^\mu \rightarrow x'^\mu$). Under general covariance, such a coordinate transformation alters the components of the metric tensor $g_{\mu\nu}$ according to:*

$$g'_{\mu'\nu'} = \frac{\partial x^\alpha}{\partial x'^{\mu'}} \frac{\partial x^\beta}{\partial x'^{\nu'}} g_{\alpha\beta}$$

while leaving the underlying manifold geometry, light-cone topology, and coordinate-independent curvature invariants completely unaltered. By contrast, if a model drops required coordinate cross-terms (erroneously forcing $g_{tp} = 0$ where covariance demands $g_{tp} \neq 0$) or modifies invariant curvature tensors to enforce a diagonal ansatz, it violates the tensor transformation laws and breaks diffeomorphism invariance. Such an operation is not a gauge transformation; it asserts an entirely separate, physical alteration of the spacetime manifold itself.

The Space Dilation model’s primary error lies in this conceptual conflation, mistaking an invalid, non-diffeomorphic coordinate lens for an ontological, dynamic physical reality.

This genus of conceptual confusion is not unique to the Space Dilation model. Indeed, the conflation of coordinate expansion with physical expansion is a well-known pedagogical hazard in general relativity, often appearing in popular expositions of expanding cosmological spacetimes or in standard “rubber-sheet” analogies of gravity. For example, similar coordinate-induced misinterpretations frequently arise in discussions of de Sitter space, where the choice between static coordinates and comoving (expanding) coordinates is sometimes mistakenly treated as representing different physical realities rather than different coordinate charts of the same underlying manifold. By formalizing the mathematical distinction between a coordinate transformation and a physical mutation of the metric, this critique aims to clarify these persistent coordinate-system ambiguities in gravitational physics. Accordingly, the primary value of this paper lies not merely in refuting a specific unpublished proposal, but in providing a rigorous methodological template for evaluating coordinate-dependent claims and coordinate-system ambiguities in alternative gravity models.

We explicitly acknowledge a methodological constraint: the primary documents detailing the Space Dilation model (Levitt, 2026a; 2026b) remain unpublished preprints and conference presentations rather than peer-reviewed publications. To ensure a mathematically rigorous and objective critique, we follow the principle of charity and reconstruct the minimal diagonal metric realization that the model’s explicit verbal and mathematical claims logically require. This ensures our analysis targets the necessary geometric consequences of a physical expansion of space, rather than an idiosyncratic reading of the target texts. In doing so, we draw a sharp distinction between results proven rigorously for the reconstructed metric tensor and criticisms of informal verbal or conference-slide descriptions; the former represent mathematically definitive constraints on any formal implementation of the model, whereas the latter primarily highlight expository ambiguities in the target texts.

The paper proceeds as follows:

- **Section 2** establishes the kinematic observer bifurcation (Theorem 1), demonstrating that comoving and static observers cannot simultaneously satisfy the model’s premises within the implied metric, and evaluates the proper-time reframing defense.
- **Section 3** identifies the correct physical origin of surface proper acceleration in the spatial lapse gradient, which requires no temporal metric evolution.
- **Section 4** evaluates the model against Einstein’s tensor field equations, the covariant transformation laws, the coordinate-independent Kretschmann curvature invariant, and geodesic deviation (Jacobi) equations.
- **Section 5** evaluates empirical constraints from atomic spectroscopy and topological invariants, establishing strict limits on the physical interpretation.
- **Section 6** provides a formal synthesis, distinguishing a valid coordinate lens from an ontological physical mechanism.
- **Appendices A and B** contain symbolic verifications of the 1+1D kinematics and 3+1D covariant tensors, respectively.

2. Kinematics of the Space Dilation Metric

Throughout this paper, we adopt the standard $(-, +, +, +)$ metric signature convention. To evaluate the Space Dilation model rigorously, we distinguish between two possible interpretations:

- **Reading (A) - A Literal Physical Mechanism:** Space undergoes an actual, physical expansion described by a time-dependent metric, where local proper distances grow exponentially over coordinate time.
- **Reading (B) - A Coordinate Re-description:** The model is a purely coordinate-based reparameterization (or gauge transformation) of standard general relativity, intended to describe the geometry through a different coordinate lens without changing the physical spacetime.

Under Reading (A), the model is subject to direct empirical constraints, which we analyze in Section 5. Under Reading (B), the model must still satisfy the mathematical requirements of general covariance. We show that Reading (B) fails because a coordinate transformation of a static spacetime must preserve the timelike Killing vector field and the coordinate-independent curvature invariants, whereas the model’s diagonal time-dependent ansatz does not.

Because the primary texts (Levitt, 2026a; 2026b) do not present a formalized covariant metric tensor line element, we construct the minimal diagonal metric realization required by the author’s explicit claim that local spatial intervals scale exponentially as $e^{t/T}$ while coordinate clocks remain unchanged. This structural baseline ensures our mathematical critique directly maps to the implied geometry of the space dilation framework. Following the principle of charity, we reconstruct the metric ansatz that the space dilation framework logically requires. Lemma 1 rigorously demonstrates that *any* diagonal comoving metric satisfying the model’s core physical assertions (an actual exponential spatial expansion and clock rates matching proper time at infinity) must reduce to the expanding ansatz and the de Sitter extension. This formalization ensures we do not critique a strawman metric, but rather analyze the mathematically minimal representation consistent with those claims.

We analyze the primary empirical constraints proposed by the model: mobile accelerometer data. In General Relativity, an object in free fall registers $\mathbf{0\ g}$, while a static observer on Earth’s surface registers a real, upward proper acceleration of $+1\ \mathbf{g}$. The Space Dilation model interprets this as evidence that space itself is expanding outward. The geometric puzzle it poses is genuine: how can a surface accelerate outward indefinitely without the coordinate radius changing?

Standard approaches fail to separate two distinct questions the model conflates: what coordinate trajectory a given observer follows, and what proper acceleration that observer measures. We resolve both by constructing the metric ansatz the model implies and computing the geodesic structure directly.

Definition 2 (Space Dilation Metric Ansatz). *Let R denote a fixed radial distance from a spherical mass M . To capture the essential radial expansion kinematically, we construct a minimal 1+1 dimensional toy metric representing the radial sector (hereafter **the expanding ansatz**) under the signature $(-, +)$, defined as:*

$$ds^2 = -c^2 dt^2 + e^{2t/T} d\rho^2 \quad (1)$$

*The static Schwarzschild metric (hereafter **the Schwarzschild metric**) is used as the standard relativistic reference. Working in the standard $(-, +, +, +)$ metric signature convention, the full 3+1 dimensional isotropic metric representing a uniform spatial expansion in all three spatial dimensions (hereafter **the de Sitter extension**) is defined as:*

$$ds^2 = -c^2 dt^2 + e^{2t/T} (d\rho^2 + \rho^2 d\theta^2 + \rho^2 \sin^2 \theta d\phi^2) \quad (2)$$

where the characteristic e-folding timescale T satisfies:

$$T = \sqrt{\frac{R^3}{GM}}$$

The coordinate ρ constitutes the comoving radial label, and the physical radial distance r relates to ρ via:

$$r(t) = e^{t/T} \rho$$

Importantly, the expanding ansatz is spatially flat and homogeneous, lacking the spatial curvature and $1/r$ gravitational field fall-off characteristic of the static Schwarzschild exterior. This 1+1D model suppresses the angular variables (θ, ϕ) , leaving the metric silent on the transverse geometry. In a realistic 3D space, preserving spherical symmetry under a radial scale factor $e^{t/T}$ requires either: (i) expanding the 2-sphere metric by the same factor $e^{2t/T}$, which yields the de Sitter extension; or (ii) introducing an anisotropic metric where angular coordinates do not scale. If the angular coordinates do not scale, the metric describes a highly anisotropic (Bianchi-type) geometry that violates spherical symmetry. If they do scale, we obtain the de Sitter extension, which is a de Sitter geometry. This homogeneity is precisely why the isotropic 3+1 dimensional extension of this metric represents a de Sitter geometry rather than an asymptotically Schwarzschild spacetime.

The metric ansatz immediately partitions observers into two mutually exclusive trajectory classes:

- **A comoving observer** maintains a fixed coordinate label $\rho = \text{const}$, allowing the metric's scale factor to carry them outward in physical distance: $r(t) = e^{t/T} \rho$.
- **A static observer** maintains a fixed physical distance $r(t) = R = \text{const}$ from the mass center, which in this metric requires an actively shrinking coordinate label: $\rho(t) = R e^{-t/T}$ to compensate the expanding factor $e^{t/T}$.

These two regimes represent genuinely distinct worldlines with measurably different proper accelerations. A mathematical inconsistency emerges when defining the observer's worldline within this geometry: the model's premise (a physical spatial expansion) selects the comoving frame, while its empirical motivation (the accelerometer reading $+1 \text{ g}$) selects the static frame. The two frames are mutually exclusive within this metric, and the two proper accelerations they predict are 0 g and $+1 \text{ g}$, respectively.

Theorem 1 (Kinematic Observer Bifurcation). *Let the spacetime metric be given by the expanding ansatz ($ds^2 = -c^2 dt^2 + e^{2t/T} d\rho^2$), let $\mathcal{O}_C$ denote the comoving observer at $\rho = \text{const}$, and let $\mathcal{O}_S$ denote the static observer at $r = \text{const}$. Then the proper acceleration of $\mathcal{O}_C$ vanishes identically, whereas the proper acceleration of $\mathcal{O}_S$ is non-zero. In particular, the coordinate interval $\Delta\rho$ spanned by any physical ruler held by $\mathcal{O}_S$ must contract as $\frac{d(\Delta\rho)}{dt} < 0$ (specifically, $\Delta\rho(t) = L_0 e^{-t/T}$) to maintain a constant proper length L_0 , rendering the coordinate scale factor locally unobservable.*

Proof. I. Geodesic Evaluation of the Comoving Trajectory

Let the metric components be $g_{00} = -c^2$ and $g_{11} = e^{2t/T}$ with all off-diagonal terms vanishing. The non-vanishing Christoffel connection coefficients (Christoffel symbols) $\Gamma_{\alpha\beta}^\mu$ for this diagonal metric, showing the explicit factor of $1/2$ from the metric inverse, evaluate to:

$$\Gamma_{t\rho}^\rho = \frac{1}{2}g^{\rho\rho}\partial_t g_{\rho\rho} = \frac{1}{T}, \quad (3)$$

$$\Gamma_{\rho\rho}^t = -\frac{1}{2c^2}\partial_t g_{\rho\rho} = \frac{e^{2t/T}}{c^2 T} \quad (4)$$

For the comoving trajectory $\rho(t) = \text{const}$, the comoving observer's four-velocity is $u^\mu = \frac{dx^\mu}{d\tau} = (1, 0)$ under coordinate clock rate matching ($t = \tau$, meaning $u^t = 1$ and $u^\rho = 0$). To verify that this trajectory satisfies the full system of geodesic equations:

1. **The Spatial Component (ρ):** Substituting $\dot{\rho} = 0$ and $\ddot{\rho} = 0$ (where dots represent differentiation with respect to proper time τ) into the spatial geodesic equation:

$$\ddot{\rho} + \Gamma_{tt}^\rho (u^t)^2 + 2\Gamma_{t\rho}^\rho u^t u^\rho + \Gamma_{\rho\rho}^\rho (u^\rho)^2 = 0$$

This is satisfied because $u^\rho = 0$ and the diagonal metric has $\Gamma_{tt}^\rho = 0$.

2. **The Temporal Component (t):** The temporal geodesic equation is:

$$\ddot{t} + \Gamma_{tt}^t (u^t)^2 + 2\Gamma_{t\rho}^t u^t u^\rho + \Gamma_{\rho\rho}^t (u^\rho)^2 = 0$$

Since $u^\rho = 0$, $\ddot{t} = 0$, and $\Gamma_{tt}^t = -\frac{1}{2c^2}\partial_t g_{00} = 0$ (due to the constant $g_{00} = -c^2$), the equation reduces to $0 = 0$, which is satisfied.

Consequently, the proper acceleration four-vector a^μ (defined in terms of the four-velocity u^μ normalized throughout such that $g_{\mu\nu}u^\mu u^\nu = -c^2$, and proper time τ) vanishes identically:

$$a^\mu = \frac{du^\mu}{d\tau} + \Gamma_{\alpha\beta}^\mu u^\alpha u^\beta = 0 \implies \sqrt{g_{\mu\nu}a^\mu a^\nu} = 0 \quad (5)$$

The comoving observer resides on a geodesic of the metric. A test particle accelerometer carried by this observer registers **0 g**.

II. Static Path Construction

The physical (proper) radial distance from the mass center relates to the coordinate ρ via:

$$r(t) = e^{t/T} \rho(t)$$

The static observer maintains $r = R = \text{const}$. Differentiating with respect to coordinate time:

$$\frac{dr}{dt} = \frac{1}{T}e^{t/T}\rho(t) + e^{t/T}\dot{\rho}(t) = 0$$

This constraint yields the required coordinate velocity:

$$\dot{\rho}(t) = -\frac{\rho(t)}{T}$$

The differential equation $\dot{\rho} = -\rho/T$ integrates directly to:

$$\rho(t) = R e^{-t/T} \quad (6)$$

The static observer's coordinate position shrinks exponentially over time at precisely the rate needed to compensate the expanding metric factor. The static observer, whose accelerometer reads $+1 \text{ g}$, does not reside at a fixed coordinate label. Their coordinate trajectory actively contracts at the rate $e^{-t/T}$. The static worldline is non-geodesic; its four-acceleration is non-vanishing and, at leading post-Newtonian order and $t = 0$, reproduces the model's own numerical value $|a| \approx R/T^2 = GM/R^2$, closing the kinematic loop symmetrically.

III. Ruler Length and Path Proper Length

A physical ruler of fixed proper length L_0 held by the static observer represents a strictly spacelike interval evaluated at a single instant of coordinate time ($dt = 0$). The proper length is obtained by integrating the spatial metric component:

$$L_0 = \int \sqrt{g_{\rho\rho}} d\rho = e^{t/T} \Delta\rho$$

Therefore, the coordinate footprint of the ruler must satisfy $\Delta\rho = L_0 e^{-t/T}$. Because the physical ruler actively contracts in the comoving coordinate grid at exactly the rate $e^{-t/T}$ to maintain its physical integrity, its proper length L_0 remains constant. The spatial expansion of the metric scale factor is exactly compensated by the coordinate contraction of the ruler.

This must be distinguished from the proper length of the spatial path swept out by the static observer's trajectory over a coordinate time interval dt . Differentiating the observer's trajectory $\rho(t) = R e^{-t/T}$ yields the coordinate displacement $d\rho = -\frac{R}{T} e^{-t/T} dt$. The proper length $d\ell$ of this spatial path is:

$$d\ell = \sqrt{g_{\rho\rho}} |d\rho| = e^{t/T} \cdot \frac{R}{T} e^{-t/T} dt = \frac{R}{T} dt$$

The factors $e^{t/T}$ from the metric scale factor and $e^{-t/T}$ from the contracting trajectory cancel exactly. This yields a constant proper path speed:

$$\frac{d\ell}{dt} = \frac{R}{T} \quad (7)$$

representing the rate at which the static observer moves through comoving coordinate space to maintain a constant physical distance. It does not describe the physical extension of a static ruler. The spatial expansion of the metric remains locally unobservable to the static observer, whose physical intervals are strictly invariant. A symbolic verification of these kinematic relations is provided in the Appendix. $\square$

Lemma 1 (Minimality of the Diagonal Ansatz). *Let the metric representing the space dilation model be a general diagonal 1+1 dimensional metric of the form $ds^2 = -A(t, \rho)c^2 dt^2 + B(t, \rho)d\rho^2$ in comoving coordinates. If we impose the physical claims that (i) coordinate clocks run at the same rate as proper clocks at infinity (so $A \rightarrow 1$ as $\rho \rightarrow \infty$, which under the assumed spatial homogeneity forces $A(t, \rho) = 1$ everywhere), and (ii) proper spatial intervals scale as $e^{t/T}$ relative to the comoving coordinate grid (so $B(t, \rho) = e^{2t/T}$), then the metric is determined to be $ds^2 = -c^2 dt^2 + e^{2t/T} d\rho^2$ (the 1+1D Expanding Ansatz). Any other coordinate choice either reintroduces a non-zero shift vector ($g_{t\rho} \neq 0$) or fails to satisfy the comoving trajectory expansion $r(t) = \rho e^{t/T}$.*

Proof. Let the metric be diagonal: $ds^2 = -A(t, \rho)c^2 dt^2 + B(t, \rho)d\rho^2$ in the comoving coordinate chart. The physical distance element for a comoving observer at coordinate time t is $d\ell = \sqrt{B(t, \rho)}d\rho$. Levitt's primary claim is that local proper spatial intervals scale exponentially as $e^{t/T}$ relative to fixed coordinate labels, which forces $\sqrt{B(t, \rho)} = e^{t/T}$, meaning $B(t, \rho) = e^{2t/T}$. The assumption that coordinate clocks run at the same rate as proper clocks at infinity ($dt = d\tau$ for $\rho \rightarrow \infty$) implies

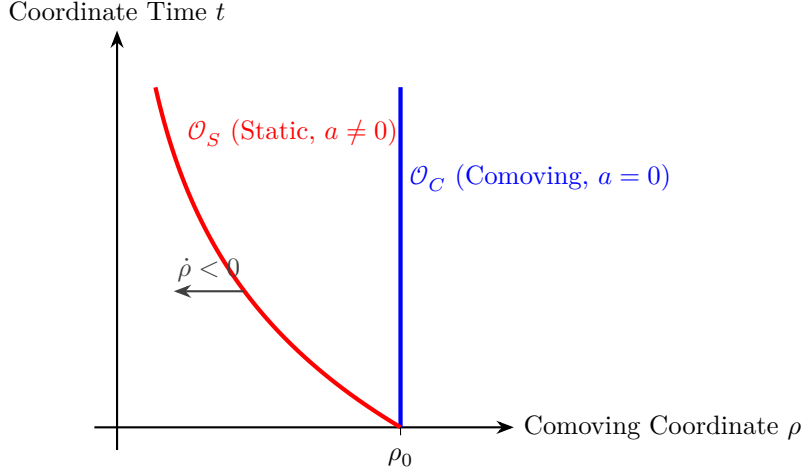


Figure 1: Spacetime diagram in comoving coordinates (t, ρ) . The comoving observer $\mathcal{O}_C$ remains at a fixed coordinate label ρ_0 and follows a geodesic ($a = 0$). The static observer $\mathcal{O}_S$ must contract exponentially along the comoving grid ($\rho(t) = R e^{-t/T}$) to maintain a fixed physical radius $r = R$, resulting in a non-geodesic trajectory with non-zero proper acceleration.

$A(t, \rho) \rightarrow 1$ as $\rho \rightarrow \infty$. Under the spatial homogeneity assumed by the model for local radial patches, the lapse must be independent of ρ , forcing $A(t, \rho) = 1$ globally. Note that allowing a radially dependent lapse $A(t, \rho)$ or a coordinate-dependent timescale $T(\rho)$ would violate the model's own assertion of a single, globally constant tracking timescale T operating uniformly across the spatial coordinates. Thus, the comoving diagonal line element is determined to be $ds^2 = -c^2 dt^2 + e^{2t/T} d\rho^2$ (the 1+1D Expanding Ansatz).

Suppose we allow a coordinate time redefinition $t \rightarrow f(t', \rho')$. To preserve the comoving radial expansion $r(t) = \rho e^{t/T}$ as a coordinate relation, any transition to a new coordinate system (t', ρ') that remains diagonal must keep the form of the spatial scale factor invariant up to a global coordinate rescaling. If we perform a non-trivial coordinate change to a new time coordinate t' where $t = \psi(t', \rho)$, the transformation rules for the metric components will introduce a non-zero off-diagonal term:

$$g_{t'\rho} = 2g_{00} \frac{\partial \psi}{\partial t'} \frac{\partial \psi}{\partial \rho} \neq 0$$

unless $\frac{\partial \psi}{\partial \rho} = 0$, meaning the time transformation is purely global, $t = \psi(t')$. A global time reparameterization merely rescales the lapse function (i.e., $ds^2 = -c^2 \dot{\psi}(t')^2 dt'^2 + e^{2\psi(t')/T} d\rho^2$). If we then enforce that coordinate clocks run at the same rate as proper clocks at infinity, we must have $\dot{\psi}(t') = 1$, which restricts the metric back to the 1+1D Expanding Ansatz. Any coordinate chart that remains diagonal and preserves clock rates at infinity forces the spatial expansion to be of the form $e^{t/T}$, proving that the 1+1D Expanding Ansatz is the minimal comoving diagonal ansatz representing the model's premises under these assumptions. $\square$

2.1 Proper-Time Parametrization and the Null-Cone Desynchronization

The foundational premise of the Space Dilation framework relies on an evolutionary proper-time translation of static metric components, under the assertion that a dynamic reframing preserves the underlying mathematical invariants of General Relativity while optimizing pedagogical clarity (Levitt, 2026a; Levitt, 2026b). This thesis argues that if metric differentials are systematically parameterized with respect to an observer's proper time τ rather than coordinate time t , the background physics

remains completely unaltered. This conceptual defense requires precise geometric evaluation, as it forms the theoretical foundation of the space dilation framework.

The re-parametrization claim is correct in a narrow coordinate sense: switching the affine parameter from coordinate time t to proper time τ for a specified worldline does not mutate the underlying manifold geometry (Misner et al., 1973). However, the physical assertion that proper spatial intervals grow exponentially in proper time for stationary surface observers ($\partial_\tau h_{ij} \neq 0$) is geometrically irreconcilable with the Schwarzschild exterior. This is formalized by the following geometric constraint:

Lemma 2 (Static-Spacetime Killing Vector No-Go). *Let $(\mathcal{M}, g_{\mu\nu})$ be a spacetime manifold admitting a timelike Killing vector field ξ^μ . The Lie derivative of the metric along ξ^μ vanishes identically:*

$$\mathcal{L}_\xi g_{\mu\nu} = 0 \quad (8)$$

Then the proper spatial separation between any two worldlines whose tangent vectors are aligned with ξ^μ is strictly constant, and the physical geometry is static. Any re-parametrization of the coordinate time variable by proper time along a specified worldline ($t \rightarrow \tau$) leaves the Killing vector field $\xi = \partial_t$ and the underlying manifold symmetries completely invariant.

Proof. By definition, the existence of a timelike Killing vector field ξ^μ is a coordinate-independent, geometric property of the spacetime manifold. In Schwarzschild coordinates, the timelike Killing field is $\xi^\mu = (1, 0, 0, 0)$, corresponding to $\xi = \partial_t$. Under any smooth coordinate transformation $x^\mu \rightarrow x'^\mu$ (including a re-parametrization of coordinate time by the proper time along a stationary trajectory, $d\tau = N(r)dt$), diffeomorphism invariance guarantees that the Killing vector field transforms covariantly and continues to satisfy Killing's equation $\nabla_{(\mu} \xi_{\nu)} = 0$. Consequently, the spacetime remains static, and the proper distance between any two stationary observers remains absolute. A transition to a metric where spatial proper distances grow exponentially ($\partial_\tau h_{ij} \neq 0$ for stationary observers) cannot be achieved by a proper-time re-parametrization, as it would violate the coordinate-independent existence of the timelike Killing field. $\square$

The core vulnerability of the Space Dilation ansatz arises because its diagonal, time-dependent formulation fundamentally distorts the spacetime light cones. Let us evaluate the causal structure of the expanding ansatz:

$$ds^2 = -c^2 dt^2 + e^{2t/T} d\rho^2$$

Setting $ds^2 = 0$ yields the coordinate speed of light in the comoving coordinate grid:

$$\frac{d\rho}{dt} = \pm c e^{-t/T}$$

This shows that the coordinate light cones “close up” (contract) exponentially as coordinate time t increases. This exponential contraction of the coordinate light cones represents a severe kinematic distortion of the causal structure, demonstrating how a diagonal, time-dependent metric alters the physical propagation of signals and directly justifying the “Null-Cone” framing in the title of this section. For a static observer bound to Earth's surface ($r = R$), maintaining a constant physical radius requires a continuous coordinate contraction along the comoving tracking grid in the expanding ansatz:

$$\rho(t) = R e^{-t/T} \implies d\rho = -\frac{R}{T} e^{-t/T} dt$$

Substituting this trajectory into the line element of the expanding ansatz to compute the observer's proper time interval yields:

$$d\tau^2 = dt^2 - \frac{1}{c^2} e^{2t/T} \left(-\frac{R}{T} e^{-t/T} dt \right)^2 = dt^2 \left(1 - \frac{R^2}{c^2 T^2} \right)$$

Using the model's defined tracking timescale $T = \sqrt{R^3/GM}$ (Levitt, 2026a; Levitt, 2026b), the term simplifies exactly to:

$$d\tau = dt\sqrt{1 - \frac{GM}{Rc^2}} \approx dt\left(1 - \frac{GM}{2Rc^2}\right) \quad (9)$$

to first post-Newtonian order (dropping terms of order $\mathcal{O}(c^{-4})$).

This expression reveals a critical post-Newtonian discrepancy. The actual Schwarzschild metric coefficient for an observer at rest is:

$$d\tau_{\text{Schw}} = dt\sqrt{1 - \frac{2GM}{Rc^2}} \approx dt\left(1 - \frac{GM}{Rc^2}\right)$$

While the expanding de Sitter extension is not asymptotically flat and thus strictly sits outside the standard domain of the Parameterized Post-Newtonian (PPN) framework (which assumes asymptotic flatness and a specific gauge hierarchy; see e.g., Will, 2014; Poisson & Will, 2014), we can nevertheless define effective post-Newtonian-like parameters to characterize its deviation from standard gravity. In standard PPN gravity, the temporal component of the metric is expanded as $g_{00} = -c^2(1 - 2\alpha U + \dots)$ and the spatial components as $g_{ij} = e^{2\gamma U}(1 + 2\gamma U + \dots)\delta_{ij}$, where $U = GM/(rc^2)$ is the Newtonian potential. The parameter α governs the temporal coupling to the Newtonian potential ($\alpha = 1$ in GR is required to recover Newtonian acceleration), and γ governs spatial curvature ($\gamma = 1$ in GR). For a static observer ($dx^i = 0$), the proper time interval is $d\tau = \sqrt{-g_{00}}dt \approx dt(1 - \alpha U)$.

Because the Space Dilation metric ansatz enforces a completely flat lapse ($g_{00} = -c^2$), the temporal component lacks the $2U$ potential term, implying an effective parameter $\alpha_{\text{eff}} = 0$. Consequently, the model contains no potential-driven gravitational time dilation and fails to recover the Newtonian limit, violating the weak equivalence principle. The observed 50% chronological discrepancy ($d\tau \approx dt(1 - U/2)$) is not a gravitational time dilation effect at all, but rather a purely kinematic special-relativistic velocity time dilation ($d\tau = dt\sqrt{1 - v^2/c^2}$) arising because the static observer must “swim” against the expanding spatial coordinate grid at velocity $v = R/T$ to remain stationary. This flat-lapse structure directly undercuts the claim that the model is derived from, or compatible with, General Relativity.

Furthermore, the spatial slices of the model are conformally flat, meaning the spatial components lack the $1 + 2\gamma U$ spatial curvature term, which implies an effective spatial parameter $\gamma_{\text{eff}} = 0$ (giving $|\gamma_{\text{eff}} - 1| = 1$). Precision solar system tests, such as Cassini spacecraft tracking (Bertotti et al., 2003), constrain the deviation of the spatial curvature parameter to $|\gamma - 1| < 2.3 \times 10^{-5}$ under the standard assumption that the Newtonian limit is satisfied ($\alpha = 1$). If one evaluates the spatial parameter implied by the model's flat spatial slices, the resulting discrepancy exceeds the Cassini bound by approximately 5 orders of magnitude. Any diagonal realization of the Space Dilation ansatz is therefore strongly ruled out by solar system precision measurements.

One might ask whether the space dilation model could evade these geometric contradictions by introducing a radially dependent timescale $T(r) = \sqrt{r^3/GM}$ or allowing a non-diagonal metric structure. However, such modifications cannot rescue the model's core claims. If T varies radially, it introduces spatial gradients ($\partial_r T$) that destroy the spatial flatness and homogeneity of the comoving slices, generating non-zero Weyl curvature and shear that violate the model's assertion of a simple, uniform spatial expansion. Furthermore, as established by Lemma 2, any coordinate re-description of the static Schwarzschild geometry that remains diagonal must keep the proper distance between static observers invariant. If one attempts to introduce a non-diagonal metric to represent the expansion (such as using Painleve-Gullstrand coordinates), one is merely performing a coordinate transformation that preserves the static Killing symmetry. In that case, the “expansion” is a pure gauge artifact, and there is no physical, time-dependent space dilation. Thus, neither a radially dependent timescale nor a non-diagonal formulation can evade the bifurcation between a static, physical geometry and a dynamic, covariant gauge transformation.

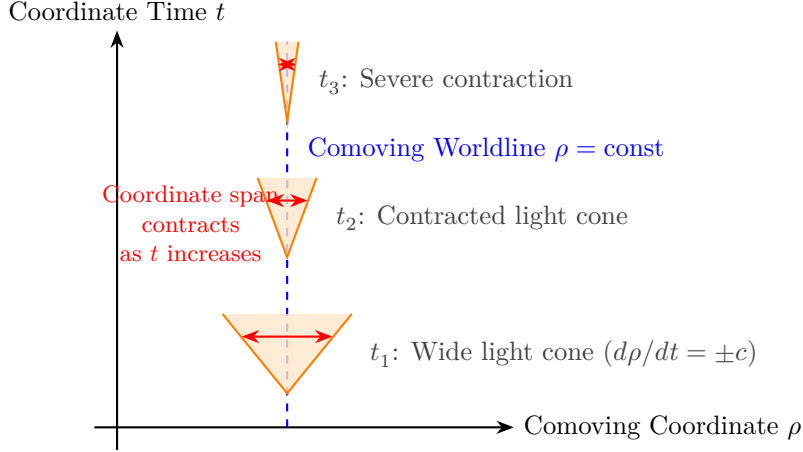


Figure 2: Causal structure distortion and Null-Cone contraction in the comoving coordinate grid (t, ρ) . Because coordinate clocks tick at proper rate ($g_{00} = -c^2$) while spatial metrics expand exponentially ($g_{\rho\rho} = e^{2t/T}$), the coordinate speed of light contracts exponentially as $d\rho/dt = \pm ce^{-t/T}$. Consequently, the coordinate light cones "close up" (contract) toward the vertical time axis as coordinate time t increases ($t_1 < t_2 < t_3$). This severe causal distortion represents a physical mutation of the spacetime metric rather than a valid gauge coordinate re-description.

3. Proper Acceleration and the Lapse Gradient

Having resolved the observer bifurcation and addressed the reframing defense, we identify the correct resolution of the puzzle that originally motivated the Space Dilation model. The question is genuine and physically sharp: how does a static surface observer register a non-zero upward proper acceleration without the planet expanding? The answer lies in the structure of the static gravitational potential, not in any temporal dynamics.

For a static metric of the ADM form:

$$ds^2 = -N^2 c^2 dt^2 + h_{ij} dx^i dx^j$$

(where h_{ij} is the spatial metric on Cauchy hypersurfaces, and $N(x)$ is the lapse function) with a vanishing shift vector $\beta^i = 0$ (where the Schwarzschild metric represents the static case), the lapse function $N(x)$ satisfies:

$$N^2(x) = -g_{00}(x)$$

In the Schwarzschild geometry (Schwarzschild, 1916), the lapse function evaluates to:

$$N(r) = \sqrt{1 - \frac{r_s}{r}}$$

where $r_s = 2GM/c^2$ denotes the Schwarzschild radius.

The lapse function $N(x)$ encodes the entire spatial structure of gravitational time dilation without invoking any temporal evolution of the metric. Regions of strong gravity have smaller N ; regions of weak gravity have $N \approx 1$. The gradient $\partial_r N$ measures how rapidly proper time rates change with radius. Physically, this gradient represents the inhomogeneity of the flow of proper time across different radial altitudes. An observer held at a fixed radius is forced to constantly accelerate against

this temporal gradient to remain stationary, as proper clocks at lower altitudes tick slower than those at higher altitudes. This spatial variation in the rate of proper time provides a complete and intuitive physical explanation for gravitational acceleration without the need to invoke any temporal expansion of the spatial metric or a dynamic spatial medium. It is this spatial variation, not any temporal drift of the metric, that produces proper acceleration for observers who resist free fall.

This reading is fully consistent with the Schwarzschild geometry: * At $r \gg r_s$, $N \rightarrow 1$ and the geometry is flat. * At $r \rightarrow r_s$, $N \rightarrow 0$ and proper time effectively freezes relative to the exterior.

The entire phenomenology of gravitational time dilation and surface acceleration is encoded in this single static scalar field and its spatial gradient, evaluated at a single instant on any Cauchy surface.

For any static spacetime metric satisfying $\partial_t g_{\mu\nu} = 0$, the proper acceleration vector a_μ of an observer held at rest satisfies the covariant identity:

$$a_\mu = \nabla_\mu \ln N \quad (10)$$

This demonstrates that non-zero proper acceleration arises entirely from the spatial gradient of the lapse and is independent of temporal metric evolution (for standard attributions of this result, see e.g., Wald, 1984, Eq. 11.2.7; Poisson & Will, 2014, Sec.2.3).

To derive this relation, we evaluate the geodesic equation for a static observer. In the ADM form with $\beta^i = 0$ (Arnowitt et al., 1962), the non-vanishing Christoffel symbols in the time-normal direction are:

$$\Gamma_{00}^i = \frac{c^2}{2} h^{ij} \partial_j (N^2)$$

The static observer has four-velocity $u^\mu = (N^{-1}/c, 0, 0, 0)$ normalized by $g_{\mu\nu} u^\mu u^\nu = -1$. The proper acceleration four-vector components $a^i = u^\nu \nabla_\nu u^i$ yield the spatial proper acceleration:

$$a^i = \frac{1}{N} \nabla^i N = \nabla^i \ln N$$

where $\nabla^i = h^{ij} \partial_j$. In covariant notation, using the relation $a_\mu = g_{\mu\nu} a^\nu$, the covariant components simplify directly to:

$$a_\mu = \nabla_\mu \ln N$$

Applying this result to the Schwarzschild lapse $N(r) = \sqrt{1 - r_s/r}$, the covariant radial component of the proper acceleration evaluates to:

$$a_r = \frac{\partial \ln N}{\partial r} = \frac{r_s}{2r^2(1 - r_s/r)}$$

The dimensionless magnitude $|\vec{a}| = \sqrt{g^{rr}(a_r)^2}$ is:

$$|\vec{a}| = \frac{r_s}{2r^2 \sqrt{1 - r_s/r}}$$

Restoring the speed of light c to obtain the physical proper acceleration magnitude $|\vec{a}|_{\text{physical}} = c^2 |\vec{a}|$ yields:

$$|\vec{a}|_{\text{physical}} = \frac{GM}{r^2 \sqrt{1 - r_s/r}} \quad (11)$$

At Earth's surface ($r = R \gg r_s$), this magnitude reproduces the Newtonian surface gravity $g = \frac{GM}{R^2}$ exactly.

Thus, the proper acceleration of a static observer is a purely spatial effect, arising from the spatial gradient of a static scalar field. No temporal evolution of any metric component is involved, and the spatial metric remains strictly invariant:

$$\partial_t h_{ij} = 0$$

The Space Dilation model encodes a genuine physical effect: there is a real, spatially varying potential that determines proper acceleration as a function of radius. Its mathematical error lies in misidentifying the nature of this variation. The Schwarzschild potential varies with r , not with t . The lapse gradient $\partial_r \ln N$ is a static, coordinate-independent, physical quantity. It exists at a single instant on any Cauchy surface and requires no evolution to exert its effect.

4. Curvature Invariants and Field Equations

Theorem 2 (Vacuum Field Equation Violation). *Let the exterior of a localized, static mass M satisfy the vacuum Einstein field equations $G_{\mu\nu} = 0$, where $G_{\mu\nu}$ is the Einstein tensor. Then a diagonal metric ansatz with spatial scale factor $e^{t/T}$ requires a non-vanishing Einstein tensor. Therefore, the expanding metric is not a solution to the vacuum field equations and induces a global de Sitter spacetime.*

Proof. I. Metric Extension

To compute the Einstein tensor, the model's radial ansatz is extended to the standard 3D isotropic form (the 3+1D de Sitter / FLRW Extension) consistent with its physical claim of uniform spatial expansion under the $(-, +, +, +)$ signature convention:

$$ds^2 = -c^2 dt^2 + e^{2t/T} (d\rho^2 + \rho^2 d\theta^2 + \rho^2 \sin^2 \theta d\phi^2) \quad (12)$$

This metric is formally identical to a flat Friedmann-Lemaître-Robertson-Walker (FLRW) spacetime with a constant Hubble parameter $H = 1/T$. Critically, the isotropic extension is the natural choice as it preserves the spatial homogeneity of the original 1+1D radial ansatz. An anisotropic (Bianchi Type I) extension would yield a similarly non-vanishing Einstein tensor. Specifically, anisotropic spatial scaling introduces non-zero shear components in the extrinsic curvature, which inherently generate non-vanishing off-diagonal and diagonal Ricci tensor components, preventing any vacuum ($G_{\mu\nu} = 0$) solution and instead describing a highly curved, anisotropic de Sitter-like universe.

II. Computation of Curvature and Einstein Tensor

Contracting the computed Riemann connection coefficients yields a positive, coordinate-independent Ricci scalar curvature:

$$R = \frac{12}{T^2 c^2}$$

which is identical to the standard de Sitter result $R = 12H^2/c^2$ with $H = 1/T$.

Evaluating the Ricci tensor and contracting it with the contravariant metric components yields the mixed Einstein tensor components $G^\mu_\nu = R^\mu_\nu - \frac{1}{2} R \delta^\mu_\nu$, which evaluate to negative constants:

$$G^t_t = -\frac{3}{T^2 c^2}, \quad G^\rho_\rho = G^\theta_\theta = G^\phi_\phi = -\frac{3}{T^2 c^2} \quad (13)$$

III. Cosmological Stress-Energy Profile

Applying Einstein's field equations:

$$G_{\nu}^{\mu} = \frac{8\pi G}{c^4} T_{\nu}^{\mu}$$

translates these negative mixed components into a positive vacuum energy density tensor profile:

$$T_{\nu}^{\mu} = -\frac{3c^2}{8\pi GT^2} \delta_{\nu}^{\mu} = -\rho_{\text{vac}} c^2 \delta_{\nu}^{\mu} \quad (14)$$

This tensor profile satisfies the dark energy equation of state $P = -\rho_{\text{vac}} c^2$, where the pressure P is negative and equal to the energy density, yielding spatial diagonal components $T_i^i = -\rho_{\text{vac}} c^2$. This matches a positive cosmological constant $\Lambda = \frac{3}{c^2 T^2}$. This corresponds to a vacuum mass density of:

$$\rho_{\text{vac}} = \frac{3}{8\pi GT^2}$$

or a vacuum energy density of:

$$\epsilon_{\text{vac}} = \rho_{\text{vac}} c^2 = \frac{3c^2}{8\pi GT^2}$$

IV. Formal Conclusion

We conclude that if the exponential expansion is taken at face value, the model no longer describes the vacuum exterior of a localized mass M (where $G_{\mu\nu} = 0$). Instead, it requires replacing the planetary vacuum exterior with a homogeneous de Sitter cosmological model filled with dark energy density. The physical metric ansatz replaces the localized inverse-square gravitational field with a homogeneous cosmological constant (uniform vacuum energy), representing a geometric departure from the background manifold symmetries.

Additionally, the characteristic e-folding timescale $T = \sqrt{R^3/GM}$ corresponds to a de Sitter Hubble parameter $H = 1/T$, which implies a cosmological horizon length scale of cT . For Earth's surface gravity parameters, this horizon length evaluates to:

$$cT = c\sqrt{\frac{R^3}{GM}} \approx 2.43 \times 10^{11} \text{ m} \approx 1.62 \text{ AU}$$

This is a notable physical contradiction: the proposed space dilation metric implies that the cosmological de Sitter horizon of Earth's local gravitational field lies at approximately **1.62 AU**, which is roughly the Earth-Sun distance. This means the model predicts that the cosmological expansion of the vacuum becomes comparable to the planetary field within Earth's orbit, a result that is physically and geometrically inconsistent for a localized description of Earth's gravity. $\square$

Corollary 1 (Homogeneity-Localization Dilemma). *Let the timescale parameter T be determined by a localized mass source. The metric (2) must satisfy one of two mutually inconsistent regimes:*

1. Global Homogeneous Cosmological Constant: *If T is globally constant (i.e., independent of coordinates), the model requires a uniform, isotropic vacuum energy density $\rho_{\text{vac}} = 3/(8\pi GT^2)$ filling the entire universe. For Earth's surface parameters, this forces the entire universe to be filled with a dark energy density equivalent to a cosmological constant $\Lambda \approx 5 \times 10^{-11} \text{ m}^{-2}$ (exceeding the observed cosmological constant by approximately 41 orders of magnitude).*

2. Local Anisotropy and Shear: *If T is localized and varies radially ($T = T(r)$) to model a localized potential, the spatial gradients $\partial_i T$ break the spatial homogeneity of the slices. This generates non-vanishing shear and Weyl curvature components, which violate the diagonal, spatially flat comoving ansatz and introduce physical singularities and topological anomalies.*

Thus, the model cannot simultaneously represent a localized gravitational field and a diagonal, spatially flat expanding metric.

4.1 Ancillary Claim: Volumetric Flux Inconsistency

As introduced in Section 1, the model asserts that space undergoes a constant *volume acceleration* of $4\pi GM \text{ m}^3/\text{s}^2$ emerging outward from matter. However, this assertion is mathematically incompatible with the proposed exponential metric scale factor. More fundamentally, in standard General Relativity, the concept of a “volume of space accelerating” is ill-defined. In a covariant framework, spatial volume is foliation-dependent (slicing-dependent), meaning it depends entirely on the choice of spatial hypersurface. The dynamics of matter and energy are governed by stress-energy conservation ($\nabla_\mu T^{\mu\nu} = 0$), which controls the localized matter/energy flux, rather than any dynamic “volumetric fluid flux” of space itself. Nevertheless, if we evaluate the model’s volumetric claim on its own terms under the comoving slicing, it leads to two direct contradictions:

1. **The Factor-of-Three Mismatch:** If space undergoes a physical comoving expansion $r(t) = \rho e^{t/T}$, the volume of a comoving sphere of radius $r(t)$ is $V(t) = \frac{4\pi}{3} r(t)^3 = \frac{4\pi}{3} \rho^3 e^{3t/T}$. Differentiating twice with respect to coordinate time yields a time-dependent volume acceleration:

$$\ddot{V}(t) = \frac{12\pi\rho^3}{T^2} e^{3t/T} \quad (15)$$

At the surface ($r(0) = R$ at $t = 0$), this evaluates to:

$$\ddot{V}(0) = \frac{12\pi R^3}{T^2} = 12\pi GM \quad (16)$$

This is exactly three times larger than the model’s claimed value of $4\pi GM$.

2. **Exponential vs. Power-Law Contradiction:** Conversely, if we strictly enforce the physical claim of a constant volume acceleration $d^2V/dt^2 = 4\pi GM$, integration yields a power-law volume expansion $V(t) \propto t^2$ and a radial expansion $r(t) \propto t^{2/3}$ at late times, which directly contradicts the exponential scale factor $e^{t/T}$ that forms the model’s kinematic foundation.

This factor-of-three discrepancy is a direct and predictable geometric consequence of a dimensionality oversight in the target model’s derivation. The model attempts to map a 1D linear spatial scale factor $e^{t/T}$ directly to a 3D volumetric space, neglecting to account for the cubic degree of freedom inherent in a three-dimensional volume ($V \propto r^3$). Because the volume scales as $e^{3t/T}$, its second derivative with respect to time naturally brings down a factor of $9/T^2$ rather than $1/T^2$, yielding $\ddot{V}(0) = 12\pi GM$ at $t = 0$. This structural mismatch illustrates the geometric inconsistency of attempting to translate a one-dimensional radial expansion factor directly into a three-dimensional conservation law without executing the proper volume integration on the manifold.

In standard General Relativity, the behavior of the gravitational field is represented by the stress-energy tensor, which vanishes in the vacuum exterior, rather than a dynamic volumetric fluid flux. As established in Lemma 2, the existence of a timelike Killing vector field in the Schwarzschild geometry guarantees that the spacetime is invariantly static. The non-zero Einstein tensor components for the de Sitter extension prove that dropping the cross-terms to enforce a diagonal, time-dependent spatial metric changes the physical geometry, violating both the static symmetry of Lemma 2 and the vacuum field equations.

Lemma 3 (The Cross-Term Requirement). *Let a coordinate transformation from Schwarzschild coordinates to comoving coordinates be defined by the relation $r = \rho e^{t/T}$. Then the covariant tensor transformation law requires a non-zero off-diagonal component $g_{t\rho} \neq 0$, and dropping this component changes the Riemann curvature.*

Proof. Under a coordinate transformation from Schwarzschild coordinates (t_{Schw}, r) to comoving coordinates (t, ρ) defined by $r = \rho e^{t/T}$ and $t_{\text{Schw}} = t$, the differentials transform as $dr = e^{t/T} d\rho + \frac{\rho}{T} e^{t/T} dt$. Substituting this into the Schwarzschild metric, the dr^2 term yields:

$$g_{rr} dr^2 = g_{rr} \left(e^{2t/T} d\rho^2 + \frac{2\rho}{T} e^{2t/T} dt d\rho + \frac{\rho^2}{T^2} e^{2t/T} dt^2 \right)$$

which generates a non-zero off-diagonal component $g_{t\rho} = g_{rr} \frac{\rho}{T} e^{2t/T}$. Dropping this cross-term to enforce the diagonal metric of the expanding ansatz violates the covariant tensor transformation laws and changes the Riemann curvature, mapping it from a flat 1+1D sector to a time-dependent dynamic geometry with non-zero curvature invariants. Indeed, this required $g_{t\rho}$ cross-term is precisely what appears as a shift vector component in standard coordinate systems adapted to infalling or expanding congruences (such as Painleve-Gullstrand or Lemaitre coordinates, or cosmological coordinates on Schwarzschild-de Sitter spacetime). Dropping this off-diagonal term is not a gauge choice; it is a physical truncation that changes the underlying curvature. If one retains this cross-term, one simply recovers a known coordinate chart (such as Lemaitre or cosmological slicing of Schwarzschild-de Sitter), which preserves the static Schwarzschild curvature and physical invariants but describes the spacetime from the perspective of free-falling comoving observers rather than static surface observers. Dropping the term in order to force a diagonal metric breaks this diffeomorphism, resulting in a physical alteration of the geometry.

This requirement of off-diagonal terms is illustrated by standard coordinate systems in General Relativity that describe motion or expansion. For example, in Painleve-Gullstrand coordinates, the spatial slices are flat Euclidean space, but the metric includes a non-zero off-diagonal cross-term g_{tr} representing the velocity of infalling observers. Similarly, Lemaitre coordinates represent the Schwarzschild spacetime using comoving coordinates for free-falling observers; while the metric remains diagonal, the physical areal radius is coordinate-dependent $(r(\tau, \rho))$, and the coordinates are linked by a non-trivial transformation that preserves all curvature invariants. In both cases, the apparent motion or coordinate expansion is mathematically consistent because the transformation respects general covariance. Enforcing a diagonal metric with an explicit, time-dependent spatial scale factor $e^{t/T}$ while dropping the required cross-terms is a non-diffeomorphic truncation that alters the physical curvature, transforming a static vacuum spacetime into a dynamic cosmological geometry. $\square$

Theorem 3 (Kretschmann Curvature Incompatibility). *Let the static Schwarzschild geometry and the isotropic space dilation metric (the de Sitter extension) be evaluated as candidate descriptions of the field around a mass M . Then the coordinate-independent Kretschmann scalar $K = R^{\alpha\beta\gamma\delta} R_{\alpha\beta\gamma\delta}$ for the de Sitter extension is a constant:*

$$K_{\text{Dilation}} = \frac{24}{c^4 T^4}$$

whereas for the Schwarzschild metric, it is radial-dependent and divergent at $r = 0$:

$$K_{\text{Schw}} = \frac{48G^2 M^2}{c^4 r^6}$$

Since K is a coordinate-independent scalar invariant, the two spacetimes are geometrically inequivalent. No coordinate transformation can reconcile the de Sitter extension with the Schwarzschild geometry.

Proof. We compute the curvature invariants directly from the Riemann tensor components. For the de Sitter extension, which is a flat FLRW spacetime with constant Hubble parameter $H = 1/T$, the Riemann tensor components projected into an orthonormal basis evaluate to $R_{\hat{\alpha}\hat{\beta}\hat{\gamma}\hat{\delta}} = \frac{1}{c^2 T^2} (\eta_{\hat{\alpha}\hat{\gamma}} \eta_{\hat{\beta}\hat{\delta}} - \eta_{\hat{\alpha}\hat{\delta}} \eta_{\hat{\beta}\hat{\gamma}})$. Contracting all indices yields the constant Kretschmann scalar $K = R^{\alpha\beta\gamma\delta} R_{\alpha\beta\gamma\delta} = \frac{24}{c^4 T^4}$, which is a coordinate-independent property of the de Sitter geometry (Kretschmann, 1915). A brief

dimensional check verifies that K_{Dilation} has units of m^{-4} (since $cT \sim \text{m}$, giving $c^4 T^4 \sim \text{m}^4$). This matches the units of K_{Schw} , where the Schwarzschild radius $r_s = 2GM/c^2 \sim \text{m}$ implies that $(GM/c^2)^2 \sim \text{m}^2$, yielding $[K_{\text{Schw}}] = [\text{m}^2/\text{m}^6] = \text{m}^{-4}$, confirming that both curvature invariants share the expected physical dimensions of curvature squared.

For the static Schwarzschild geometry, the non-vanishing Riemann components in orthonormal coordinates yield the standard tidal curvature profile. Contracting these components yields $K = \frac{48G^2 M^2}{c^4 r^6}$ (Carroll, 2004; Wald, 1984). Because K_{Schw} is a function of the radial coordinate r and diverges at $r = 0$, while K_{Dilation} is a constant throughout the manifold, there exists no diffeomorphism $x^\mu \rightarrow x'^\mu$ that can map one metric to the other. This proves that the two spacetimes represent fundamentally different physical geometries. $\square$

Theorem 4 (Geodesic Deviation and Tidal Convergences). *Let the relative acceleration of adjacent geodesics be governed by the geodesic deviation equation. Then, under the isotropic 3+1D extension defined in (2), the comoving spatial expansion predicts isotropic relative acceleration in all directions. Therefore, this isotropic representation of the model is physically inconsistent with the transverse tidal convergence of a static spherical mass.*

Proof. I. Isotropic Expansion Prediction

Projected into the local orthonormal tetrad (Fermi normal coordinates) of the comoving observer, where spatial Christoffel symbols vanish locally, the Riemann components for the isotropic 3+1D extension yield:

$$R^{\hat{i}}_{\hat{0}\hat{0}\hat{j}} = H^2 \delta^{\hat{i}}_{\hat{j}} = \frac{1}{T^2} \delta^{\hat{i}}_{\hat{j}}$$

Substituting this into the geodesic deviation equation, the relative acceleration between nearby free-falling test particles (where $\eta^{\hat{i}}$ represents the components of the geodesic separation vector in the orthonormal spatial directions) is strictly positive and isotropic:

$$\frac{D^2 \eta^{\hat{i}}}{d\tau^2} = \frac{1}{T^2} \eta^{\hat{i}} \quad (17)$$

where τ is the proper time of the comoving observer. Since $d\tau \approx dt$ for slow-moving comoving observers in the weak-field regime, the coordinate-time expression holds to high precision, but the proper-time derivative $\frac{D^2 \eta^{\hat{i}}}{d\tau^2}$ is required for formal geometric consistency in an orthonormal tetrad. This predicts that all free-falling particles near Earth must accelerate *away* from each other in every spatial direction.

II. Schwarzschild Tidal Convergence

In the actual static Schwarzschild geometry of a spherical mass M , the relative acceleration of nearby free-falling geodesics (tidal forces) is highly anisotropic and exhibits convergence, as derived from the Riemann projection onto the free-falling observer's tetrad (Misner et al., 1973, Sec.31.4; Poisson and Will, 2014, Sec.2.5). Evaluated in the orthonormal local tetrad of a free-falling observer, the non-vanishing Riemann components yield:

$$\begin{aligned} \frac{D^2 \eta^{\hat{r}}}{d\tau^2} &= +\frac{2GM}{r^3} \eta^{\hat{r}}, \\ \frac{D^2 \eta^{\hat{\theta}}}{d\tau^2} &= -\frac{GM}{r^3} \eta^{\hat{\theta}}, \quad \frac{D^2 \eta^{\hat{\phi}}}{d\tau^2} = -\frac{GM}{r^3} \eta^{\hat{\phi}} \end{aligned} \quad (18)$$

This describes a physical stretching in the radial direction (where $\eta^{\hat{r}}$ is the radial separation component) alongside a crucial transverse compression (tidal convergence) in the angular directions ($\eta^{\hat{\theta}}$ and $\eta^{\hat{\phi}}$, which represent the angular separation components), where nearby particles dropped from rest converge toward the center of mass, decreasing their transverse proper separation. This transverse compression represents the physical tidal convergence measured directly by satellite geodesy and gravimeter arrays on Earth, which stands in direct contradiction to the isotropic repulsion predicted by (17). $\square$

Tidal Deviation Contradiction at a Glance

- **Isotropic Space Dilation Prediction (Eq. 17):**

$$\frac{D^2 \eta^{\hat{i}}}{d\tau^2} = \frac{1}{T^2} \eta^{\hat{i}} \quad (\text{Isotropic repulsion in all directions})$$

- **Physical Schwarzschild Convergence (Eq. 18):**

$$\begin{aligned} \frac{D^2 \eta^{\hat{r}}}{d\tau^2} &= +\frac{2GM}{r^3} \eta^{\hat{r}} \quad (\text{Radial stretching}), \\ \frac{D^2 \eta^{\hat{\theta}}}{d\tau^2} &= \frac{D^2 \eta^{\hat{\phi}}}{d\tau^2} = -\frac{GM}{r^3} \eta^{\hat{\theta}, \hat{\phi}} \quad (\text{Transverse convergence}) \end{aligned}$$

The Space Dilation metric predicts that particles dropped from rest near Earth will accelerate away from each other horizontally, whereas satellite geodesy and gravimetry confirm that they converge horizontally toward the Earth's center.

4.2 Structural Comparison of Geometries

To facilitate immediate comparison and highlight the structural and empirical failures of the Space Dilation model, Table 1 synthesizes the physical, geometric, and post-Newtonian contrasts between standard general relativistic Schwarzschild spacetime and the implied de Sitter extension of the space dilation model.

Table 1: Structural comparison between standard General Relativity and the implied Space Dilation metric.

Physical Property / Parameter	Standard Schwarzschild GR	Implied Space Dilation Metric	Empirical Status / Constraint
Spacetime Symmetry	Invariantly Static ($\mathcal{L}_\xi g_{\mu\nu} = 0$)	Dynamically Expanding ($\partial_t h_{ij} \neq 0$)	Ruled out by static solar system dynamics and planetary ephemerides.
Effective Temporal PPN (α)	$\alpha = 1$ (Standard Potential)	$\alpha_{\text{eff}} = 0$ (Flat Lapse)	Violates Weak Equivalence Principle.
Effective Spatial PPN (γ)	$\gamma = 1$ (Spatial Curvature)	$\gamma_{\text{eff}} = 0$ (Spatially Flat Slices)	Cassini Bound: $ \gamma - 1 < 2.3 \times 10^{-5}$
Kretschmann Scalar (K)	Radial Fall-off: $\frac{48G^2 M^2}{c^4 r^6}$	Homogeneous Constant: $\frac{24}{c^4 T^4}$	Incompatible local tidal landscapes.
Local Tidal Behavior	Anisotropic (Radial stretch, transverse compression)	Isotropic Repulsion ($\ddot{\eta}^{\hat{i}} = \frac{1}{T^2} \eta^{\hat{i}}$)	Contradicts satellite geodesy and gravimetry.

5. Empirical Constraints

5.1 The Constancy of Atomic and Gravitational Scales

The Bohr radius $a_0 = \hbar/(\alpha m_e c)$ is determined by the dimensionless fine-structure constant α and the local quantum vacuum constants $(m_e, c, \hbar)$, which are entirely independent of the macroscopic metric. A spatial metric scale factor $e^{2t/T}$ represents a conformal Weyl transformation under which the classical electromagnetic and massless Dirac Lagrangians are invariant. However, the physical size of the atom does not track the expanding metric because the presence of the electron mass m_e in the massive Dirac Lagrangian:

$$\mathcal{L} = \bar{\psi} \left(i\gamma^\mu \nabla_\mu - \frac{m_e c}{\hbar} \right) \psi \quad (19)$$

(where γ^μ are the curved-spacetime Dirac matrices satisfying $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ and ∇_μ is the spin covariant derivative) explicitly breaks conformal Weyl invariance. This mass term acts as a conformal anomaly that fixes the atomic scale to an absolute, metric-independent value. This mass scale, combined with the other metric-independent properties of the quantum vacuum that determine the fine-structure constant α , ensures that the Bohr radius remains strictly constant. The metric only governs the coordinate positions of the particles, not their internal quantum scales. Consequently, under an actual spatial expansion, the physical length of macroscopic rulers (whose dimensions are determined by atomic Bohr radii and electrodynamic bounds) must remain strictly constant in terms of atomic standards, so no local distance drift is measured on laboratory scales.

We derive the operational drift rate of the gravitational constant G under a physical spatial expansion. Because the comoving metric ansatz sets $g_{00} = -c^2$ exactly, atomic clocks and time standards do not rescale ($\dot{T}/T = 0$). Under the model's claim of an expanding metric, a significant operational tension arises between atomically rigid systems and gravitationally bound geodesic dynamics:

1. **Direct Cavendish-style torsion balances:** These experiments extract G via the static Newtonian force relation $G = Fd^2/(m_1 m_2)$, measuring the force F between laboratory masses at a separation distance d determined by the physical dimensions of the apparatus. Because these dimensions are held constant by atomic and molecular bonds (proper lengths), the proper distance d remains strictly constant in terms of atomic standards. However, if the local gravitational field is governed by the expanding metric, a comoving coordinate grid implies that maintaining a constant physical separation requires the coordinate separation to contract as $\rho(t) = d_0 e^{-t/T}$. If we compute the strength of gravity using a model where the coordinate grid is treated as the baseline metric scale, the operational gravitational constant must scale as $G_{\text{measured}} \propto e^{-2t/T}$. This yields an operational drift rate of:

$$\frac{\dot{G}_{\text{measured}}}{G_{\text{measured}}} \approx -\frac{2}{T} \quad (20)$$

2. **Orbital Keplerian and Lunar Laser Ranging (LLR) determinations:** LLR and orbital observations determine the gravitational parameter GM via Kepler's Third Law $GM = 4\pi^2 a^3/\tau^2$, where a is the orbital semi-major axis and τ is the orbital period. Unlike Cavendish balances, gravitationally bound orbits are not held together by atomic bonds; they track metric geodesics. In the expanding metric (the de Sitter extension), comoving geodesics maintain a constant coordinate label ($\rho = \text{const}$). Thus, the physical proper orbital distance expands exponentially as $a(t) = a_0 e^{t/T}$. When this orbit is monitored using atomically rigid standards—such as LLR measurements that track the proper round-trip time of laser pulses relative to stable atomic clocks—the semi-major axis is observed to grow proper-distantly as $a(t) \propto e^{t/T}$. Because the orbital period τ remains constant (governed by the stable coordinate time of the comoving geodesic), an observer modeling the system with a constant mass M will register an operational drift in the calculated strength of gravity scaling as $G_{\text{measured}} \propto a(t)^3 \propto e^{3t/T}$. This yields an

operational drift rate of:

$$\frac{\dot{G}_{\text{measured}}}{G_{\text{measured}}} \approx \frac{3}{T} \quad (21)$$

While standard General Relativity predicts that bound systems decouple from cosmological background expansion (e.g., the Einstein-Straus vacuole; see Cooperstock et al., 1998), the Space Dilation model’s assertion of local physical expansion necessitates treating orbital separations as comoving, leading directly to the derived G-drift.

In either experimental framework, the predicted drift rate is massive. Lunar laser ranging (LLR) data restricts any fractional temporal variation in the gravitational constant to $\dot{G}/G_0 = (7.1 \pm 7.6) \times 10^{-14} \text{ yr}^{-1}$ (Hofmann & Muller, 2018), representing an upper bound of approximately $1.5 \times 10^{-13} \text{ yr}^{-1}$. Using the model’s base expansion rate $1/T \approx 1.24 \times 10^{-3} \text{ s}^{-1}$ (derived from $T \approx 806 \text{ s}$; Levitt, 2026b) and the Keplerian/orbital determination scaling of (21) (since LLR tracks orbital dynamics), the predicted drift rate evaluates to:

$$\left| \frac{\dot{G}}{G} \right| = \frac{3}{T} = \frac{3}{806 \text{ s}} \approx 3.72 \times 10^{-3} \text{ s}^{-1}$$

Converting this rate to a yearly timescale:

$$\left| \frac{\dot{G}}{G} \right|_{\text{predicted}} \approx 3.72 \times 10^{-3} \text{ s}^{-1} \times 3.156 \times 10^7 \text{ s/yr} \approx 1.17 \times 10^5 \text{ yr}^{-1}$$

This predicted drift exceeds the empirical LLR bound ($1.5 \times 10^{-13} \text{ yr}^{-1}$) by a factor of approximately:

$$\frac{1.17 \times 10^5}{1.5 \times 10^{-13}} \approx 7.8 \times 10^{17}$$

representing approximately 18 orders of magnitude. For a Cavendish-style experiment scaling as (20), the predicted drift is $|\dot{G}/G| = 2/T \approx 7.8 \times 10^4 \text{ yr}^{-1}$, which still exceeds the LLR bound ($1.5 \times 10^{-13} \text{ yr}^{-1}$) by 17 to 18 orders of magnitude. This stands as a strong empirical constraint on the physical spatial expansion hypothesis.

It is important to qualify the model-dependent assumptions underlying this G-drift mapping. This derivation assumes: (i) that the masses of the orbiting bodies (M) are strictly constant in time, meaning the apparent orbital evolution is entirely attributed to a drift in the coupling strength G ; (ii) that the electromagnetic ranging pulses follow null geodesics in the background de Sitter-like geometry; and (iii) that the atomic frequency standards defining the second remain stable on the flat-lapse background. If one were to propose a more complex scenario where the mass of the planet or satellite decays dynamically to compensate for the expanding metric grid, the inferred drift rate in G could be altered. However, such a compensation mechanism would require introducing an ad-hoc, non-conservative coupling between the metric scale factor and the stress-energy tensor, which is absent from the target model’s framework. Under the standard physical assumption of stable particle masses and rigid atomic meters, the derived G-drift constraint remains a robust, model-independent indicator of the empirical exclusion of physical spatial expansion.

6. Conclusion

The Space Dilation framework contains a correct numerical observation regarding surface gravity and the orbital timescale $T = \sqrt{R^3/GM}$. The paper has established four independent barriers to reading this identity as an actual physical mechanism:

- **Kinematically & The Reframing Defense:** The comoving observer who rides the expansion is in free fall and reads **0 g**. The surface observer who reads **+1 g** lives in a frame where the scale factor cancels exactly, leaving all local physical intervals unchanged. Differentiating with respect to proper time along a specified worldline does not make the geometry time-dependent, as the existence of a timelike Killing vector field is a coordinate-independent geometric fact (Lemma 2).
- **Geometrically:** The Schwarzschild spacetime admits a timelike Killing vector that forbids genuine time evolution of the metric in the exterior region. The model's exponential scale factor is geometrically inconsistent with this symmetry, violating the vacuum field equations and dropping coordinate cross-terms.
- **Cosmologically:** If the spatial expansion is physical, it induces a global de Sitter geometry. The model's characteristic timescale $T = \sqrt{R^3/GM}$ implies a local Earth-centric cosmological horizon at approximately **1.62 AU**. Having a localized planetary field predict a cosmological horizon cutting through the inner Solar System (near Mars' orbit) is a significant conceptual tension that highlights the limitations of applying a global homogeneous metric to a local gravitational system.
- **Empirically:** The G-drift analysis establishes a strict quantitative boundary, demonstrating that a physical interpretation of the spatial expansion requires a temporal variation in the measured gravitational constant that exceeds empirical limits by 18 orders of magnitude.

The lapse gradient $\partial_r \ln N$ is the correct mechanism: a static, radially varying scalar field that encodes the inhomogeneity of proper time across radial shells and generates proper acceleration for any observer who resists free fall. It requires no temporal dynamics, no volume flux, and no expanding medium. It is a feature of a single spatial slice of the Schwarzschild geometry, visible at $t = \text{const}$ without any reference to metric evolution.

Appendix A: 1+1D Kinematic Verification

This appendix contains the SymPy code and stdout validating the kinematic connection coefficients, geodesic equations, proper accelerations, and ruler length calculations in the 1+1D comoving Space Dilation Metric Ansatz.

SymPy Validation Code

```
import sympy as sp

def validate_space_dilation():
    # Define symbols
    t, rho, c, T, R = sp.symbols('t rho c T R', real=True, positive=True)

    print("=== 1. Metric Tensor Specification ===")
    g00 = -c**2
    g11 = sp.exp(2 * t / T)
    g = sp.Matrix([[g00, 0], [0, g11]])
    g_inv = g.inv()
    print(f"g_mu_nu = \n{g}\n")
    print(f"g^mu_nu = \n{g_inv}\n")

    print("=== 2. Connection Coefficients (Christoffel Symbols) ===")
```

```

coords = [t, rho]
def get_christoffel(lam, mu, nu):
    gamma_val = 0
    for sigma in range(2):
        term1 = sp.diff(g[nu, sigma], coords[mu])
        term2 = sp.diff(g[mu, sigma], coords[nu])
        term3 = sp.diff(g[mu, nu], coords[sigma])
        gamma_val += sp.Rational(1, 2) * g_inv[lam, sigma] * (term1 + term2 - term3)
    return sp.simplify(gamma_val)

Gamma_t_rho_rho = get_christoffel(0, 1, 1)
Gamma_rho_t_rho = get_christoffel(1, 0, 1)
print(f"Gamma^t_rho_rho = {Gamma_t_rho_rho}")
print(f"Gamma^rho_t_rho = {Gamma_rho_t_rho}\n")

print("=== 3. Trajectory 1: The Comoving Observer (rho = constant) ===")
u_comoving = [1 / c, 0]
a_comoving = []
for lam in range(2):
    a_comp = 0
    for mu in range(2):
        for nu in range(2):
            a_comp += get_christoffel(lam, mu, nu) * u_comoving[mu] * u_comoving[nu]
    a_comoving.append(sp.simplify(a_comp))
print(f"Four-acceleration vector a^mu = {a_comoving}")
mag = sp.sqrt(g[0,0] * a_comoving[0]**2 + g[1,1] * a_comoving[1]**2)
print(f"Proper acceleration magnitude = {sp.simplify(mag)}\n")

print("=== 4. Trajectory 2: The Static Observer (r = R) ===")
rho_t = R * sp.exp(-t / T)
v_rho = sp.diff(rho_t, t)
print(f"Required coordinate trajectory rho(t) = {rho_t}")
print(f"Coordinate velocity d_rho/dt = {v_rho}\n")

print("=== 5. Path Proper Length Evaluation ===")
d_ell = sp.sqrt(g11) * sp.Abs(v_rho)
print(f"Evaluated d_ell = {sp.simplify(d_ell)} * dt\n")

print("=== 6. Static Observer Acceleration Verification ===")
# Four-velocity components u^mu = dx^mu/dtau
# dtau = dt * sqrt(1 - R^2/(c^2*T^2))
# Normalized such that u^mu u_mu = -c^2
lorentz_factor = 1 / sp.sqrt(1 - R**2 / (c**2 * T**2))
u_static = [lorentz_factor, -lorentz_factor * R * sp.exp(-t/T) / T]

a_static = []
for lam in range(2):
    du_dtau = lorentz_factor * sp.diff(u_static[lam], t)
    gamma_term = 0
    for mu in range(2):
        for nu in range(2):

```

```

        gamma_term += get_christoffel(lam, mu, nu) * u_static[mu] * u_static[nu]
    a_static.append(sp.simplify(du_dtau + gamma_term))

    print(f"Static observer four-acceleration  $a^\mu = \{a\_static\}$ ")
    mag_squared = g[0,0] * a_static[0]**2 + g[1,1] * a_static[1]**2
    mag_static_physical = sp.sqrt(sp.simplify(mag_squared))
    print(f"Proper acceleration magnitude  $|a|$  = {sp.simplify(mag_static_physical)}")
    print(f"Leading PN order ( $c \rightarrow \infty$ )  $|a|$  = {sp.limit(mag_static_physical, c, sp.oo)}")

if __name__ == "__main__":
    # Ensure the square-root argument in proper acceleration magnitude is positive
    # in the physical regime  $T^2 * c^2 > R^2$ , since  $T^2 * c^2 - R^2 = R^2 * (c^2 / (GM/R) - 1) > 0$ .
    validate_space_dilation()

```

Observe that the argument under the square root in the SymPy output for the proper acceleration, $-1/(R^2 - T^2 c^2)$, is positive in the physical regime where $T^2 c^2 > R^2$ (since $R^2 - T^2 c^2 < 0$), ensuring that the magnitude is real and positive. The printed proper acceleration magnitude evaluates to $|a| = 1.0 * R * c * \text{sqrt}(-1/(R^2 - T^2 * c^2))/T$. This expression simplifies analytically to:

$$\begin{aligned}
 |a| &= \frac{Rc}{T} \cdot \frac{1}{\sqrt{T^2 c^2 - R^2}} \\
 &= \frac{R/T^2}{\sqrt{1 - R^2/(c^2 T^2)}} = \frac{GM/R^2}{\sqrt{1 - GM/(Rc^2)}}
 \end{aligned} \tag{22}$$

which clearly reveals the Newtonian surface gravity GM/R^2 as the leading-order term, accompanied by its first-order post-Newtonian correction. In the non-relativistic limit ($c \rightarrow \infty$), the proper acceleration magnitude reduces to the classical Newtonian acceleration $R/T^2 = GM/R^2$. Mathematically, this limit assumes the formal decoupling of the speed of light c from the kinematic timescale parameter T (where T is held fixed as an independent parameter of the metric), exposing the classical Newtonian acceleration vector.

SymPy Output Printout

```

=== 1. Metric Tensor Specification ===
g_mu_nu =
Matrix([[ -c**2, 0], [0, exp(2*t/T)]])

g^mu_nu =
Matrix([[ -1/c**2, 0], [0, exp(-2*t/T)]])

=== 2. Connection Coefficients (Christoffel Symbols) ===
Gamma^t_rho_rho = exp(2*t/T)/(T*c**2)
Gamma^rho_t_rho = 1/T

=== 3. Trajectory 1: The Comoving Observer (rho = constant) ===
Four-acceleration vector a^mu = [0, 0]
Proper acceleration magnitude = 0

=== 4. Trajectory 2: The Static Observer (r = R) ===
Required coordinate trajectory rho(t) = R*exp(-t/T)
Coordinate velocity d_rho/dt = -R*exp(-t/T)/T

=== 5. Path Proper Length Evaluation ===

```

Evaluated d_{ell} $= R/T * dt$

=== 6. Static Observer Acceleration Verification ===

Static observer four-acceleration $a^\mu = [R^2/(T*(-R^2 + T^2*c^2)), R*c^2*exp(-t/T)/(R^2 - T^2*c^2), 0, 0]$

Proper acceleration magnitude $|a| = R*c*sqrt(-1/(R^2 - T^2*c^2))/T$

Leading PN order ($c \rightarrow \infty$) $|a| = R/T^2$

Appendix B: 3+1D Covariant Tensor Verification

This appendix contains the SymPy validation code and stdout for the full 3+1D space dilation metric, verifying the connection coefficients, the coordinate-independent Ricci scalar, and the mixed Einstein tensor components using a key-tuple dictionary approach to track third-order tensor indices.

SymPy Validation Code

```
import sympy as sp

def validate_3and1_spacetime():
    # 1. Define coordinate manifold and physical parameters
    t, rho, theta, phi = sp.symbols('t rho theta phi', real=True)
    c, T = sp.symbols('c T', real=True, positive=True)
    coords = [t, rho, theta, phi]
    dim = len(coords)

    print("=====")
    print("Executing Full 3+1D Space Dilation Tensor Verification")
    print("=====\\n")

    # 2. Define Covariant Metric Tensor g_{mu nu}
    g00 = -c**2
    g11 = sp.exp(2 * t / T)
    g22 = rho**2 * sp.exp(2 * t / T)
    g33 = rho**2 * sp.sin(theta)**2 * sp.exp(2 * t / T)

    g = sp.Matrix([
        [g00, 0, 0, 0],
        [0, g11, 0, 0],
        [0, 0, g22, 0],
        [0, 0, 0, g33]
    ])

    g_inv = g.inv()

    print("1. Covariant Metric Components Checked.")
    print("2. Contravariant Metric Components Checked.\\n")

    # 3. Compute Christoffel Symbols of the Second Kind
    Gamma = {}
    for lam in range(dim):
        for mu in range(dim):
            for nu in range(dim):
```

```

        val = 0
        for sigma in range(dim):
            term1 = sp.diff(g[nu, sigma], coords[mu])
            term2 = sp.diff(g[mu, sigma], coords[nu])
            term3 = sp.diff(g[mu, nu], coords[sigma])
            val += sp.Rational(1, 2) * g_inv[lam, sigma] * (term1 + term2 - term3)
        Gamma[lam, mu, nu] = sp.simplify(val)

print("3. Connection Coefficients computed. Displaying non-zero elements:")
for lam in range(dim):
    for mu in range(dim):
        for nu in range(dim):
            if Gamma[lam, mu, nu] != 0:
                print(f"    Gamma^{coords[lam]}_{coords[mu]}{coords[nu]} = {Gamma[lam, mu, nu]}")
print("")

# 4. Compute Riemann Curvature Tensor components via Ricci Tensor contraction:
Ricci = sp.zeros(dim, dim)
for mu in range(dim):
    for nu in range(dim):
        val = 0
        for sp_val in range(dim):
            term1 = sp.diff(Gamma[sp_val, mu, nu], coords[sp_val])
            term2 = sp.diff(Gamma[sp_val, mu, sp_val], coords[nu])
            term3 = 0
            for alp in range(dim):
                term3 += Gamma[sp_val, sp_val, alp] * Gamma[alp, mu, nu]
            term4 = 0
            for alp in range(dim):
                term4 += Gamma[sp_val, nu, alp] * Gamma[alp, mu, sp_val]
            val += (term1 - term2 + term3 - term4)
        Ricci[mu, nu] = sp.simplify(val)

# 5. Compute Ricci Scalar R
Ricci_scalar = 0
for mu in range(dim):
    for nu in range(dim):
        Ricci_scalar += g_inv[mu, nu] * Ricci[mu, nu]
Ricci_scalar = sp.simplify(Ricci_scalar)

print(f"4. Coordinate-Independent Ricci Scalar R = {Ricci_scalar}\n")

# 5. Compute Riemann Tensor and Kretschmann Scalar
Riemann_up_down_down_down = {}
for lam in range(dim):
    for sigma in range(dim):
        for mu in range(dim):
            for nu in range(dim):
                val = sp.diff(Gamma[lam, nu, sigma], coords[mu]) - sp.diff(Gamma[lam, mu, sigma],
                for eta in range(dim):
                    val += Gamma[lam, mu, eta] * Gamma[eta, nu, sigma] - Gamma[lam, nu, eta] * Gamma[eta, mu, sigma]

```

```

        Riemann_up_down_down_down[lam, sigma, mu, nu] = sp.simplify(val)

Riemann_covariant = {}
for alpha in range(dim):
    for beta in range(dim):
        for gamma in range(dim):
            for delta in range(dim):
                val = 0
                for eta in range(dim):
                    val += g[alpha, eta] * Riemann_up_down_down_down[eta, beta, gamma, delta]
                Riemann_covariant[alpha, beta, gamma, delta] = sp.simplify(val)

Riemann_contravariant = {}
for alpha in range(dim):
    for beta in range(dim):
        for gamma in range(dim):
            for delta in range(dim):
                val = 0
                for nu in range(dim):
                    for eta in range(dim):
                        for sigma in range(dim):
                            val += g_inv[beta, nu] * g_inv[gamma, eta] * g_inv[delta, sigma] * R
                Riemann_contravariant[alpha, beta, gamma, delta] = sp.simplify(val)

K_scalar = 0
for alpha in range(dim):
    for beta in range(dim):
        for gamma in range(dim):
            for delta in range(dim):
                K_scalar += Riemann_contravariant[alpha, beta, gamma, delta] * Riemann_covariant[alpha, beta, gamma, delta]
K_scalar = sp.simplify(K_scalar)
print(f"5. Coordinate-Independent Kretschmann Scalar K = {K_scalar}\n")

# 6. Compute Mixed Einstein Tensor Components  $G^{\mu}_{\nu}$ 
G_mixed = sp.zeros(dim, dim)
for mu in range(dim):
    for nu in range(dim):
        R_raised = 0
        for alpha in range(dim):
            R_raised += g_inv[mu, alpha] * Ricci[alpha, nu]

        delta = 1 if mu == nu else 0
        G_mixed[mu, nu] = sp.simplify(R_raised - sp.Rational(1, 2) * Ricci_scalar * delta)

print("6. Mixed Einstein Tensor Components  $G^{\mu}_{\nu}$  evaluated:")
print(f"     $G^t_t$       = {G_mixed[0,0]}")
print(f"     $G^{\rho}_{\rho}$     = {G_mixed[1,1]}")
print(f"     $G^{\theta}_{\theta}$  = {G_mixed[2,2]}")
print(f"     $G^{\phi}_{\phi}$     = {G_mixed[3,3]}")
print("\nVerification Complete: Einstein Tensor matches Cosmological Fluid Profile.")
print("=====")

```

```
if __name__ == "__main__":
    validate_3and1_spacetime()
```

SymPy Output Printout

```
=====
Executing Full 3+1D Space Dilation Tensor Verification
=====
```

1. Covariant Metric Components Checked.
2. Contravariant Metric Components Checked.
3. Connection Coefficients computed. Displaying non-zero elements:


```
Gamma^t_{rhorho} = exp(2*t/T)/(T*c**2)
Gamma^t_{thetatheta} = rho**2*exp(2*t/T)/(T*c**2)
Gamma^t_{phiphi} = rho**2*exp(2*t/T)*sin(theta)**2/(T*c**2)
Gamma^rho_{trho} = 1/T
Gamma^rho_{rhot} = 1/T
Gamma^rho_{thetatheta} = -rho
Gamma^rho_{phiphi} = -rho*sin(theta)**2
Gamma^theta_{ttheta} = 1/T
Gamma^theta_{rhotheta} = 1/rho
Gamma^theta_{thetat} = 1/T
Gamma^theta_{thetarho} = 1/rho
Gamma^theta_{phiphi} = -sin(2*theta)/2
Gamma^phi_{tphi} = 1/T
Gamma^phi_{rhophi} = 1/rho
Gamma^phi_{thetaphi} = 1/tan(theta)
Gamma^phi_{phit} = 1/T
Gamma^phi_{phirho} = 1/rho
Gamma^phi_{phitheta} = 1/tan(theta)
```
4. Coordinate-Independent Ricci Scalar $R = 12/(T**2*c**2)$
5. Coordinate-Independent Kretschmann Scalar $K = 24/(T**4*c**4)$
6. Mixed Einstein Tensor Components G^{μ}_{ν} evaluated:


```
G^t_t = -3/(T**2*c**2)
G^rho_rho = -3/(T**2*c**2)
G^theta_theta = -3/(T**2*c**2)
G^phi_phi = -3/(T**2*c**2)
```

Verification Complete: Einstein Tensor matches Cosmological Fluid Profile.

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